pfi

Python for Fixed Income

COURSE 2 · BOND MATH

The Price-Yield Relationship

WEEK 1 · DAY 3

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Above, at, and below 100

Day 2's four prices, each at its own yield, on 30 September 2026

Issuer	Coupon %	Yield %	Price
Pellwick Paper	4.75	4.6	100.4159
Quorvane Energy	5.25	5.4	99.3503
Dunmarrow Rail	6.125	6.125	100.0000
Halvering Foods	7.50	7.7	98.6227

The price is a function of the yield. When the yield moves, the price moves the other way.

Today: why a bond prices above, at, or below 100; how far each price moves; the pull to par; and a yield restated at another frequency.

Week 1: price and yield

Day 1	Time value of money
Day 2	Cash flows and price from yield
Day 3	**The price-yield relationship**
Day 4	Yield from price
Day 5	Case study: a new issue calendar

Par, premium, and discount

On a coupon date	Price	Here
coupon above yield	premium	Pellwick
coupon equal to yield	par, 100	Dunmarrow
coupon below yield	discount	Quorvane, Halvering

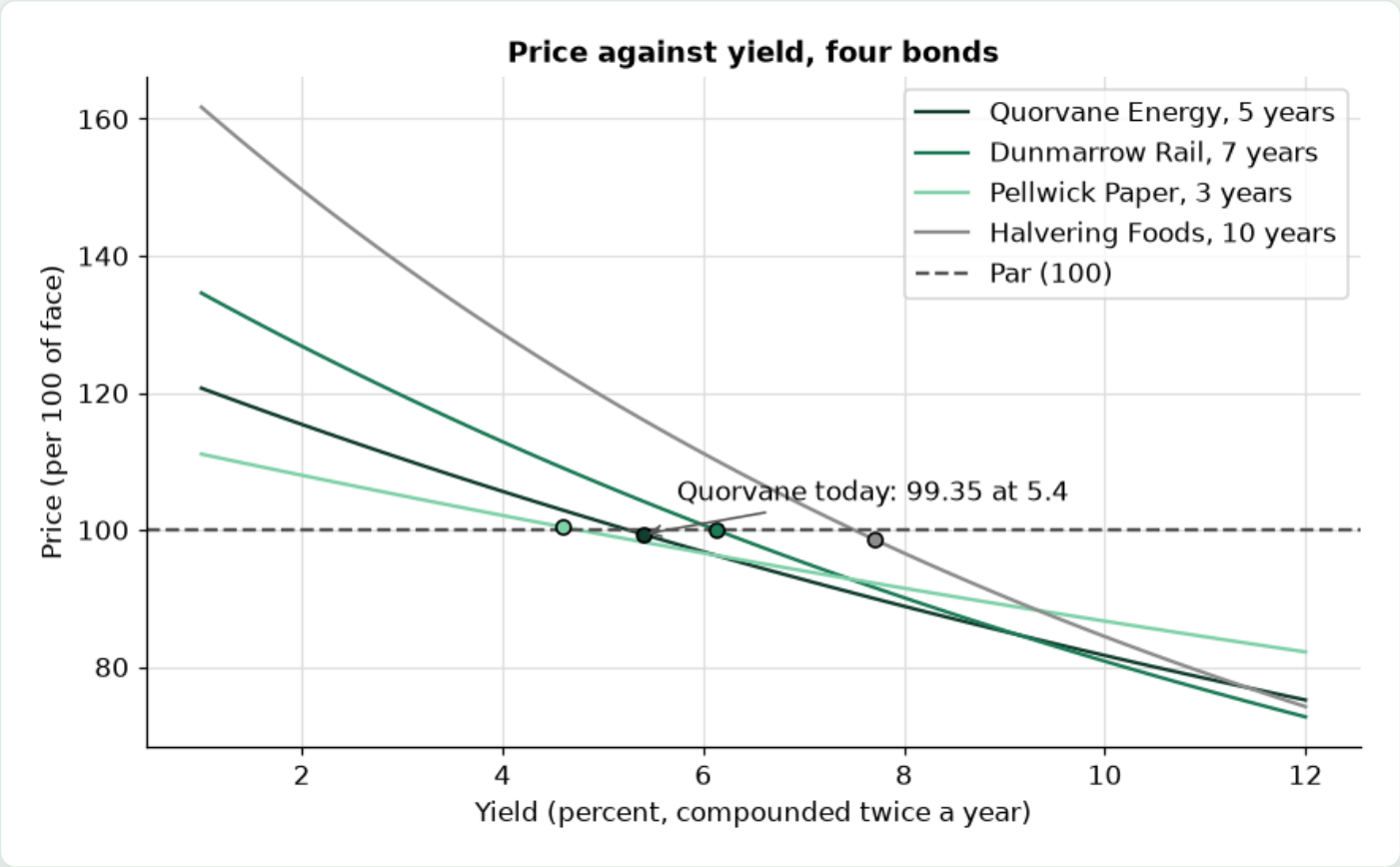
Coupon equal to yield prices at 100 for any maturity: each coupon pays exactly the yield on the 100 lent.

A premium is the present value of the coupon's excess over the yield. A discount is the present value of its shortfall.



Pellwick: 0.075 per half-year, six times. Halvering: 0.100 short, twenty times.

The price-yield curve



+4.4276

Quorvane, yield 100 bp lower

-4.2055

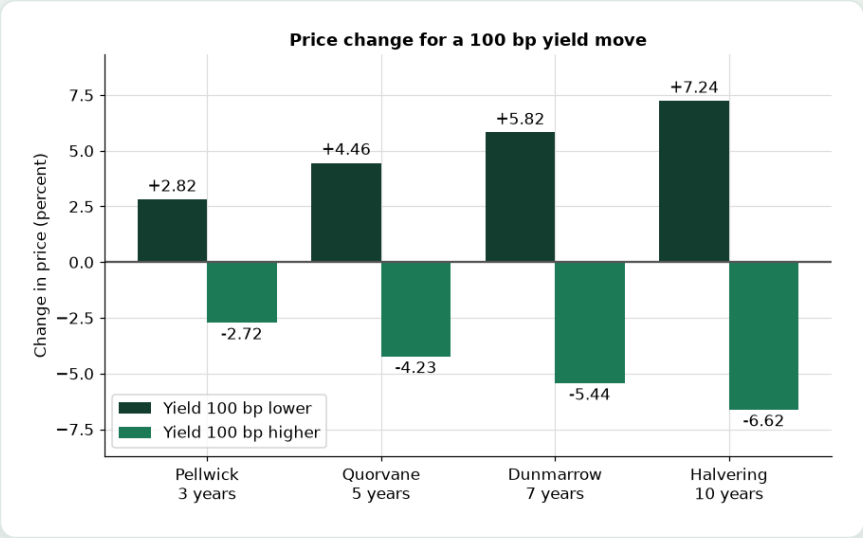
Quorvane, yield 100 bp higher

Price and yield move in opposite directions, and the curve bends.

The longest bond, Halvering, is the steepest; the shortest, Pellwick, the flattest.

Each line crosses 100 where the yield equals its coupon. Day 10 names the bend: convexity.

Which bond moves more: yield up 100 bp



Maturity and coupon tables: what-ifs on Quorvane's terms, 5.25 percent coupon, 5.40 percent yield.

Maturity alone

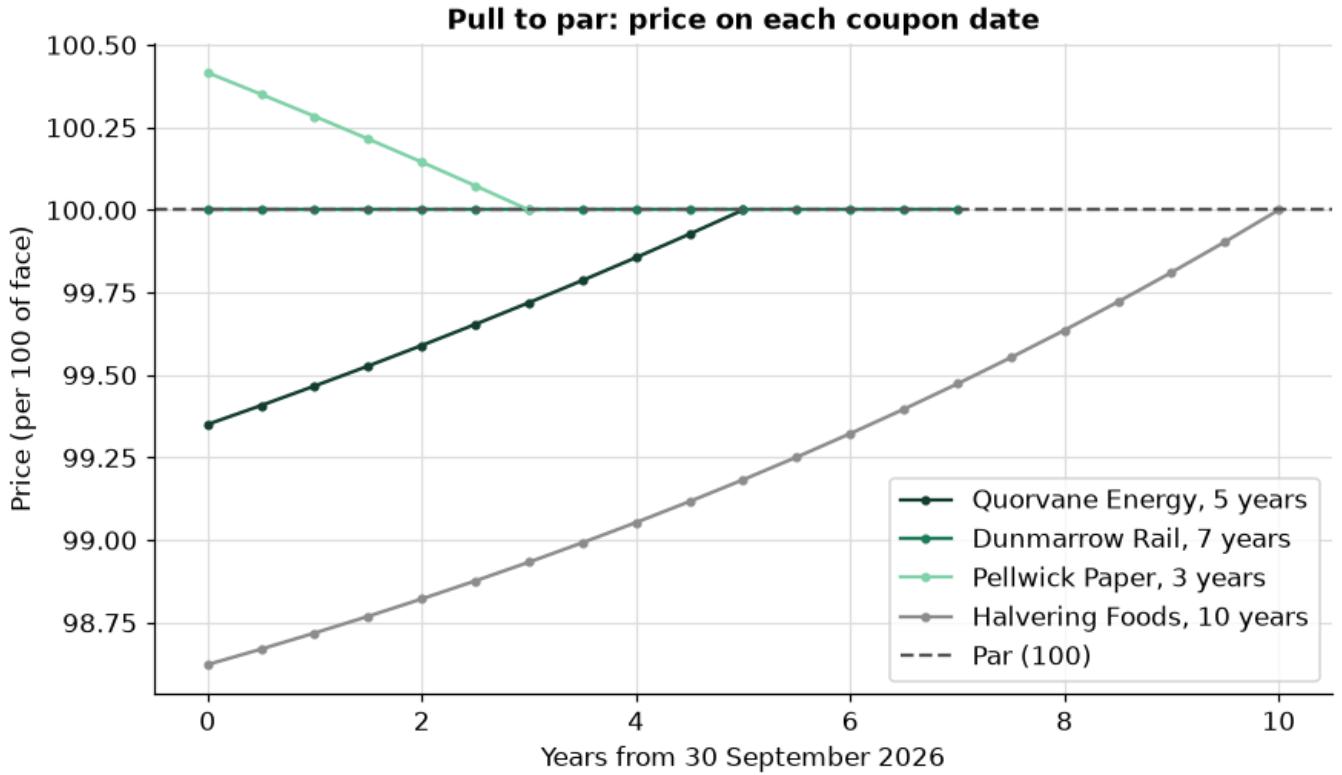
Years	Change %
1	-0.95
3	-2.70
5	-4.23
10	-7.34
20	-11.26
30	-13.33

Coupon alone, 5 years

Coupon %	Change %
0	-4.74
2.5	-4.47
5.25	-4.23
8	-4.05
10.5	-3.91

Longer maturity and lower coupon both push the weight of the cash flows later, so the price moves more. Day 9 measures it: duration.

Pull to par: the yield never moves



98.6227

Halvering today, at 7.70 percent

99.9037

Halvering with half a year left

Premium bonds drift down to 100, discount bonds drift up, a par bond stays at 100.

Halvering, next half-year: 3.85 percent, half of 7.70

If the yield does not change, the bond earns its yield.

Restating a yield: same growth in one year

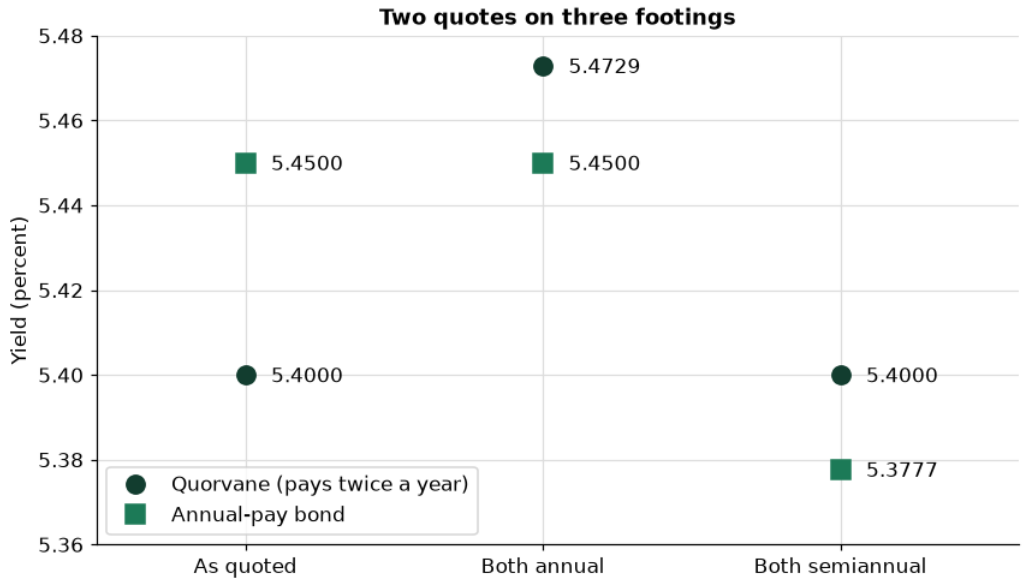
```
100 x (1 + 5.40/100/2)^2 = 105.4729
```

Compounded	Quorvane's yield %
once a year	5.4729
twice a year	5.4000
four times a year	5.3640

```
# convert_yield: one year's growth, then the new frequency
growth_one_year = (
    1 + yield_pct / 100 / from_frequency) ** from_frequency
period_rate = growth_one_year ** (1 / to_frequency) - 1
return period_rate * to_frequency * 100
```

Excel side by side (tested in Excel)

`=EFFECT(5.4/100,2)`	0.054729
`=NOMINAL(6.25/100,2)`	0.061553
`=NOMINAL(A1,4)*100`	5.364034, EFFECT in A1



Today's Excel, every formula tested

Job	Excel formula (tested)	Returns
Premium or discount	`=PRICE(DATE(2026,9,30),DATE(2029,9,30),4.75/100,4.6/100,100,2,0)`	100.415887
Price-yield column	`=PRICE(DATE(2026,9,30),DATE(2031,9,30),5.25/100,A2/100,100,2,0)`	113.35 to 87.35
The chart	`select A1:B8, Insert, Scatter with Straight Lines`	one line
100 bp up	`=PRICE(DATE(2026,9,30),DATE(2036,9,30),7.5/100,8.7/100,100,2,0)`	92.092797
Pull to par	`=-PV(7.7/100/2,2*A1,7.5/2,100), years left in A`	98.62 to 99.90
To annual	`=EFFECT(5.4/100,2)`	0.054729
To semiannual	`=NOMINAL(6.25/100,2)`	0.061553

Excel takes decimals: every formula writes out the /100, and a rate comes back as a decimal.

A message that says what and why

Part	Says	Today
Subject	what changed	Add convert_yield, tested against Excel EFFECT and NOMINAL
Body	why	yields at other frequencies must be restated to compare

```
git commit -m "<subject>" -m "<body>"
git log --oneline
git show --stat HEAD

    Add convert_yield, tested against Excel ...
 bondmath.py      | 19 +++++++
 test_bondmath.py | 18 ++++++
```

Write the subject as an instruction: it completes the sentence "This commit will ...". git log --oneline shows only the subject.

In six months, git log is how you find the change that matters.

Bring it home: paste, test, read, commit

1

Your folder

Set-Location to my_bondmath, in your projects folder.

2

Paste

Today's three append cells at the end of your two files, without the %%writefile line.

3

Test

python -m pytest -q: 8 passed, or 10 with the exercises of days 1 and 2.

4

Commit

git diff --stat, git add, then git commit with a subject and a body.

5

Read

git log --oneline and git show --stat HEAD. Press q to leave a pager.

Pasting at the end keeps anything of your own. Copy fallen behind? Take the start-of-day files from today's setup cell, and git diff shows what changed.

In Colab: download bondmath.py and test_bondmath.py from the work folder again. They replace yesterday's downloads.

The AI check

The docstring is the prompt. The code that comes back runs, and has one convention error.

1

Docstring

Units, convention, and the Excel formula: what you asked for.

2

Code back

Written for the course in the style of an AI answer. Read it.

3

Test first

Assert against Excel's reference rows. Run it: it fails.

4

Fix

Find the convention it broke, fix it, run the test again.

Today: convert_yield uses the percent figure as a decimal, and the round-trip test still passes.

Code that runs is not the same as code that is right.

What you can do now

- Say from coupon and yield whether a bond is at a premium, par, or a discount
- Draw the price-yield curve in matplotlib and Excel, and say why it bends
- Say which bond moves more: the longer one, and the lower coupon
- Show the pull to par, and that a bond earns its yield if its yield holds
- Restate a yield with convert_yield, EFFECT, and NOMINAL; commit what and why

Week 1, Day 4 Yield from price: Goal Seek, a bracket halved by hand, and a loop